# ELE458 Topics and Formulas for Exam 1 Spring 2026

## Topics

1. Relationship between pole locations and settling time
2. Design of analog compensator for position control of 2nd-order plant
3. Digital state-feedback regulator design.
   (a) Choice of sampling interval T.
   (b) Choice of `spoles` using all of the applicable methods discussed in class.
   (c) Calculation of the feedback gain vector using Matlab `place` function.
4. Interconnected transfer functions to overall state-space model.
5. Differential equations to overall state-space model.

## Formulas

1. Given a negative unity feedback system with $G(s)$ in the forward path, the transfer function of the closed-loop system is $G(s)/[1 + G(s)]$.

2. The settling time corresponding to a real-valued pole at $-p$ or complex-conjugate poles with real part $-p$ is given by $T_S = 4.6/p$. For repeated real poles at $-p$ (two poles located at $-p$), the settling time formula is $T_S = 1.45 * 4.6/p$. The settling time of a stable control system is given approximately by the largest settling time of any of its poles.

3. The recommended sampling interval $T$ for a digital state-feedback regulator is the smaller of $\frac{Ts}{20n}$ and $\frac{\pi}{5\beta_{max}}$ where $n$ is the order of the plant model, $T_S$ is the desired settling time, and $\beta_{max}$ is the maximum imaginary part of the (analog) plant poles. The largest acceptable value of $T$ is the smaller of $\frac{Ts}{2n}$ and $\frac{\pi}{5\beta_{max}}$.

4. A general $n$-th order transfer function from input $u(t)$ to output y(t),

$$\frac{b_0 s^n + b_1 s^{n-1} + \cdots + b_n}{s^n + a_1 s^{n-1} + \cdots + a_n},$$

has the following OCF state-space model: $\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t)$, $y(t) = \mathbf{C}\mathbf{x}(t) + Du(t)$, where

$$\mathbf{A} = \begin{bmatrix} -a_1 & 1 & 0 & \cdots & 0 \\ -a_2 & 0 & 1 & \cdots & 0 \\ -a_3 & 0 & 0 & \ddots & 0 \\ \vdots & \vdots & \vdots & \cdots & 1 \\ -a_n & 0 & 0 & \cdots & 0 \end{bmatrix}, \ \mathbf{B} = \begin{bmatrix} b_1 - a_1 b_0 \\ b_2 - a_2 b_0 \\ b_3 - a_3 b_0 \\ \vdots \\ b_n - a_n b_0 \end{bmatrix}, \ \mathbf{C} = [1 \ 0 \ \cdots \ 0], \ D = b_0.$$